

Ordered-Pair Symmetry Quotients of Exact Finite First-Passage Laws

Anonymous

Abstract

We quotient the 400 ordered pairs in a frozen twenty-target first-passage model by its faithful order-1920 label action. The ambient lifted order 11520 is retained as a distinct quantity. Reconstructing the induced action from all 65536 support indicators gives 272 ordered-pair orbits, with size spectrum $144 \times 1 + 128 \times 2$. The complete exact joint-survival and mixed-moment law is constant on every orbit. Orbit-stabilizer, a finite Burnside count, relation types, and the transpose involution are independently certified. All results are finite combinatorics under NO_BAD_EULER_OR_ROOT_NUMBER.

1 Frozen model and action

Let L be the sixteen named labels and let $H_0, \dots, H_{19}$ be the actual subgroup targets. A uniform permutation π of L has prefix support A_k , and

$$T_i(\pi) = \min\{k : H_i \leq \Phi(A_k)\}.$$

The frozen source supplies every hit indicator on 2^L and all joint survival counts

$$J_{ij}(k, \ell) = \#\{\pi : T_i(\pi) > k, T_j(\pi) > \ell\}.$$

Five named label permutations generate a faithful group G of order 1920. Its lifted source group has order 11520; the two orders are never identified. Transport of closures induces $gH_i = H_{gi}$ and the diagonal pair action $g(i, j) = (gi, gj)$.

Lemma 1 (Joint-law transport). *For every $g \in G$ and every ordered pair (i, j) ,*

$$J_{ij}(k, \ell) = J_{gi, gj}(k, \ell) \quad (0 \leq k, \ell \leq 16).$$

Consequently all C90 mixed raw moments and covariances are constant on the diagonal G -orbit of (i, j) .

Proof. Relabeling a permutation is a bijection and carries each prefix support A to gA . Generator-level verification on all support indicators proves $H_i \leq \Phi(A)$ exactly when $H_{gi} \leq \Phi(gA)$. Hence $T_i(\pi) = T_{gi}(g\pi)$ and likewise for j . Counting the joint events proves the identity. Mixed moments and covariance are finite functions of the joint law. $\square$

2 Pair-orbit theorem

Theorem 1. *The diagonal action of G on the 400 ordered target pairs has 272 orbits. Their exact statistics are*

<i>orbit size 1</i>	<i>144</i>	<i>stabilizer order 1920</i>	<i>144</i>
<i>orbit size 2</i>	<i>128</i>	<i>stabilizer order 960</i>	<i>128</i>
<i>diagonal</i>	<i>16</i>	<i>incomparable</i>	<i>132</i>
<i>forward-comparable</i>	<i>62</i>	<i>reverse-comparable</i>	<i>62</i>

Pair transposition induces an involution with 20 fixed orbits. Both coordinate projections recover the sixteen single-target orbits.

Proof. Direct enumeration partitions all 400 pairs. Every orbit size divides $|G|$, giving the displayed stabilizers. Target inclusion is equivariant, so each orbit has one relation type. Transposition commutes with the diagonal action, hence descends to an involution. Finally, the fixed-pair sum

$$\sum_{g \in G} |\text{Fix}_{\{H_i\}}(g)|^2 = 522240 = 1920 \cdot 272$$

independently verifies the orbit count by Burnside's lemma. $\square$

3 Independent certificate

The producer decodes an exact closure as the unique maximal-order hit target. The checker instead matches the twenty-bit hit vector of every support to a column of the subgroup-inclusion matrix. These independent decoders yield the same target maps and all 272 rows. Each row stores its complete pair membership, stabilizer, relation type, transpose index, covariance, and a SHA-256 digest of the full C90 joint payload. SymPy checks every orbit–stabilizer equation and the fixed-point quotient. Clean replay fixes the evidence digest

099d8f32794d6967b3f2653f92dcaa0b096c711b67ed070330d7763a146bc696.

All 14 hostile mutations are rejected.

4 Scope

The invocation of Burnside’s lemma is only a finite orbit count. We do not construct or claim a full Burnside ring or table of marks. No arithmetic/local data, Euler factors, root numbers, automorphy, or Hilbert–Polya operator is claimed.